

Alexander Wilson

CONTACT INFORMATION	awlego@gmail.com linkedin.com/in/alexanderwaynewilson	Bay Area, CA github.com/awlego
OBJECTIVE	Build the kind of change I want to see in the world, learning something along the way.	
EDUCATION	California Institute of Technology , Pasadena, CA B.S., Electrical Engineering June 2014	
EXPERIENCE	Patenttext , Founding Engineer 2025 – Current - Designed and built major Patenttext features: invention graph creation, detailed description, claim drafting agents, patent figure generation, novel concept discovery. - Engineering of the following varieties: RAG, eval, tool, workflow, safety, costs, latency, fine-tuning, agent, multi-agent, and observability. - Vastly improved LLM's ability to comprehend novel inventions via guided creation of to build deeply structured understanding. Structure withheld upgrades throughout GPT4 to 5.6, continuing to deliver super-LLM gains. Independent AI Research and Consulting 2022 – 2025 - Machine Learning Researcher - Replicated interpretability research of DeepMind's chess AI knowledge visualization paper. My work may be found here. - Consultant, UCLA Department of Geology - Supported graduate student research by teaching both ML fundamentals and advising on modern approaches. Kuna Systems 2019 – 2021 - Senior Machine Learning Engineer - Tech lead of the AI/ML team training deep neural nets for object detection. - Reduced missed detections by 90%; enabled a new low cost product via quantization. BioIntelliSense (formerly Striiv) 2015 – 2019 - Machine Learning Engineer - Trained heart arrhythmia detection algorithms using random forest models and xgboost. Designed data pipelines for the training. - Firmware Engineer - Wrote embedded software for small ARM-based systems interfacing with a variety of sensors and peripherals. - Designed electronics hardware for a mass produced fitness tracker with over a month of battery life. SpaceX Summers 2012 – 2014 - Electrical Engineer Intern - Avionics Remote Input/Output Team - Engineered circuits and PCBs for critical avionics sensor systems, achieving optimal weight and power efficiency. Custom Circuit Entrepreneur 2011 – 2012 Developed and marketed RGB laser driver circuits - Created, tested, and sold programmable laser drivers with 30+ effects, managing all aspects of design, testing, and sales.	
SKILLS	Languages: TypeScript, Python LLM / Agents: Vercel AI SDK, tool calling, structured outputs, LLM-as-judge evals ML: PyTorch, TensorFlow, spaCy, interpretability, RL, computer vision Infra: Cloudflare Workflows, Terraform, GCP, AWS, Postgres, FastAPI	